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1. (a) $\left\{\begin{bmatrix}1\\0\\0\end{bmatrix},\begin{bmatrix}1\\0\\1\end{bmatrix},\begin{bmatrix}1\\1\\1\end{bmatrix}\right\} \subseteq R^3$

(b) $\left\{\begin{bmatrix}1\\0\\1\end{bmatrix},\begin{bmatrix}0\\0\\0\end{bmatrix},\begin{bmatrix}0\\1\\0\end{bmatrix}\right\}$ is not linearly independent ∴ not the basis

(c) $\left\{\begin{bmatrix}1\\0\\-2\end{bmatrix},\begin{bmatrix}3\\2\\-4\end{bmatrix},\begin{bmatrix}-3\\-5\\1\end{bmatrix}\right\}$ $\because \begin{bmatrix}-3\\-5\\1\end{bmatrix} = \frac{9}{2}\begin{bmatrix}1\\0\\-2\end{bmatrix} - \frac{5}{2}\begin{bmatrix}3\\2\\-4\end{bmatrix}$

∵ not linearly independent ∴ not the basis

2. $\begin{bmatrix}-4\\3\\1\\2\end{bmatrix} = -2\begin{bmatrix}1\\0\\-2\\1\end{bmatrix} + \begin{bmatrix}-2\\3\\-3\\4\end{bmatrix}$

$\therefore S_{max} = \left\{\begin{bmatrix}1\\0\\-2\\1\end{bmatrix},\begin{bmatrix}-2\\3\\-3\\4\end{bmatrix},\begin{bmatrix}3\\-6\\1\\-3\end{bmatrix},\begin{bmatrix}-5\\9\\-4\\-7\end{bmatrix}\right\}$

3. (a) $Am = X$

$$\left[\begin{array}{cc|c}-1&3&2\\2&4&6\\3&2&6\end{array}\right] \to \left[\begin{array}{cc|c}-1&3&2\\0&10&10\\0&11&12\end{array}\right] \to \left[\begin{array}{cc|c}1&0&1\\0&1&1\\0&0&1\end{array}\right]$$

∵ Is inconsistent ∴ $x \notin \text{span}(X_1, X_2)$

(b) If $y \in \text{Span}(X_1, X_2)$ $\begin{bmatrix}-9\\-2\\5\end{bmatrix} = a\begin{bmatrix}-1\\2\\3\end{bmatrix} + b\begin{bmatrix}3\\4\\2\end{bmatrix} \to \begin{bmatrix}-1&3\\2&4\\3&2\end{bmatrix}\begin{bmatrix}a\\b\end{bmatrix} = \begin{bmatrix}-9\\-2\\5\end{bmatrix}$

$$\left[\begin{array}{cc|c}-1&3&-9\\2&4&-2\\3&2&5\end{array}\right] \to \left[\begin{array}{cc|c}-1&3&-9\\0&10&-20\\0&11&-22\end{array}\right] \to \left[\begin{array}{cc|c}-1&3&-9\\0&1&-2\\0&1&-2\end{array}\right] \to \left[\begin{array}{cc|c}1&0&3\\0&1&-2\\0&0&0\end{array}\right]$$

∵ consistent ∴ $y \in \text{Span}(X_1, X_2)$

4. $S = \{(x,y,z)^T \mid xy = z^2\}$

(1) $0 \in S$

(2) $\because (\alpha X)\cdot(\alpha y) = \alpha^2 xy$ $(\alpha z)^2 = \alpha^2 z^2$ $\therefore (\alpha x)(\alpha y) = (\alpha z)^2$ ∴ $\forall u \in S$, $\alpha u \in S$

$\because x_1 y_1 = z_1^2$ $x_2 y_2 = z_2^2$ $\therefore x_1 y_1 + x_2 y_2 = z_1^2 + z_2^2$

(3) let $[x_1, y_1, z_1]^T$, $[x_2, y_2, z_2]^T \in S$

$x_1 y_1 = z_1^2$, $x_2 y_2 = z_2^2$

$\therefore x_1 y_1 + x_2 y_2 = z_1^2 + z_2^2$

$(x_1 + x_2)(y_1 + y_2) = x_1 y_1 + x_1 y_2 + x_2 y_1 + x_2 y_2$

$(z_1 + z_2)^2 = z_1^2 + 2z_1 z_2 + z_2^2 = x_1 y_1 + 2z_1 z_2 + x_2 y_2$

$\because x_1 y_2 + x_2 y_1$ not always equal to $2z_1 z_2$

$\therefore S = \{(x,y,z)^T \mid xy = z^2\}$ is not a subspace of R^3.

5. $\begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ -3 & 0 & 6 & -1 \\ 3 & 4 & -2 & 1 \\ 2 & 0 & -4 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 3 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 9 & 9 & 8 \\ 0 & 4 & 4 & 0 \\ 0 & -6 & -6 & -4 \end{bmatrix} \to \begin{bmatrix} \boxed{1} & 3 & 1 & 3 \\ 0 & \boxed{1} & 1 & 0 \\ 0 & 0 & 0 & \boxed{8} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\therefore$ The basis for $V = \left\{ \begin{bmatrix} 1 \\ 0 \\ -3 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 0 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ -1 \\ 1 \\ 2 \end{bmatrix} \right\}$

6. S is linearly independent

$\because x_1+x_2+x_3+x_4=0 \quad \therefore x_4 = -(x_1+x_2+x_3)$

$\therefore X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ -(x_1+x_2+x_3) \end{bmatrix}$

$X = x_1\begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} + x_2\begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} + x_3\begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ -(x_1+x_2+x_3) \end{bmatrix} \quad \therefore \dim(U) = 3$

7. (a) $\dim(U)=3$ (b) $\dim(U)=3$ (c) $\dim(U)=2$ (d) $\dim(U)=2$

8. $B = \begin{bmatrix} 7 & -3 \\ 5 & -1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & -2 \\ -5 & 2 \end{bmatrix}$

transition from C to B: $B^{-1}C = \frac{1}{8}\begin{bmatrix} -1 & 3 \\ -5 & 7 \end{bmatrix}\begin{bmatrix} 1 & -2 \\ -5 & 2 \end{bmatrix} = \frac{1}{8}\begin{bmatrix} -16 & 8 \\ -40 & 24 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ -5 & 3 \end{bmatrix}$

transition from B to C $C^{-1}B = -\frac{1}{8}\begin{bmatrix} 2 & 2 \\ 5 & 1 \end{bmatrix}\begin{bmatrix} 7 & -3 \\ 5 & -1 \end{bmatrix} = -\frac{1}{8}\begin{bmatrix} 24 & -8 \\ 40 & -16 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}$

9. (a) proof: (1) $\left[\begin{array}{ccc|c} 1 & -3 & 3 & 0 \\ 0 & 4 & -6 & 0 \\ 0 & 0 & 3 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$

$\therefore Bx = 0 \quad x = 0 \quad \therefore$ B is linearly independent

(2) $X = (x_1+\frac{3}{4}x_2+\frac{1}{2}x_3)\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (\frac{1}{4}x_2+\frac{1}{2}x_3)\begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix} + \frac{1}{3}x_3\begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix}$

$\therefore$ Thus $\text{span}(B) = R^3$

(b) $\because x_1 = -8, \ x_2 = 2, \ x_3 = 3$

$\therefore X = (-8+\frac{3}{2}+\frac{3}{2})\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + (\frac{1}{2}+\frac{3}{2})\begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix} + 1\begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix} = -5\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2\begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix}$

(c) $y = 1\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2\begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -10 \\ 9 \end{bmatrix}$

10. (a) $\left[\begin{array}{cc|c} 3 & 6 & 1 \\ 1 & 2 & 1 \end{array}\right] \to \left[\begin{array}{cc|c} 3 & 6 & 1 \\ 0 & 0 & \frac{2}{3} \end{bmatrix}$ $\because$ inconsistent $\therefore b \notin \text{col}(A)$

(b) $\left[\begin{array}{cc|c} 0 & 1 & 2 \\ 1 & 0 & 5 \\ 0 & 1 & 2 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 0 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right]$ $\because$ consistent $\therefore b \in \text{col}(A)$

(c) $\left[\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & 4 & 10 \\ 1 & 2 & 5 \end{array}\right] \to \left[\begin{array}{cc|c} 1 & 2 & 5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$ $\because$ consistent $\therefore b \in \text{col}(A)$

11. (a) $A=\begin{bmatrix}1&2&3&5&0&2&4\\2&1&3&4&7&8&9\\1&1&2&3&2&2&5\end{bmatrix}\to\begin{bmatrix}1&2&3&5&0&2&4\\0&-3&-3&-6&7&4&1\\0&-1&-1&-2&2&0&1\end{bmatrix}\to\begin{bmatrix}\boxed{1}&2&3&5&0&2&4\\0&\boxed{-3}&-3&-6&7&4&1\\0&0&0&0&\boxed{\frac{1}{3}}&\frac{4}{3}&-\frac{1}{3}\end{bmatrix}$

$\to\begin{bmatrix}1&0&1&0&-14&14&\\0&1&1&2&0&8&-5\\0&0&0&0&1&4&-2\end{bmatrix}$

The basis for $col(A)=\left\{\begin{bmatrix}1\\2\\1\end{bmatrix},\begin{bmatrix}2\\1\\1\end{bmatrix},\begin{bmatrix}0\\2\\2\end{bmatrix}\right\}$

(b) $Ax=0$

solution set: $\begin{cases}x_1=-x_3-x_4+14x_6-14x_7\\x_2=-x_3-2x_4-8x_6+5x_7\\x_5=-4x_6+2x_7\end{cases}$

$Null(A)=span\left\{\begin{bmatrix}-1\\-1\\1\\0\\0\\0\\0\end{bmatrix},\begin{bmatrix}-1\\-2\\0\\1\\0\\0\\0\end{bmatrix},\begin{bmatrix}14\\8\\0\\-4\\1\\0\\0\end{bmatrix},\begin{bmatrix}-14\\5\\0\\2\\0\\0\\1\end{bmatrix}\right\}=span(u)$

$\dim(Null(A))=4$

12. $\because y_3=y_1+y_2=x_2-x_1+x_3-x_2=x_3-x_1$

$\therefore y_1, y_2, y_3$ are not linearly independent

13. $A=\begin{bmatrix}1&2&2&3&1&4\\2&4&5&5&4&9\\3&6&7&8&5&9\end{bmatrix}\to\begin{bmatrix}1&2&2&3&1&4\\0&0&1&-1&2&1\\0&0&1&-1&2&-3\end{bmatrix}\to\begin{bmatrix}1&2&2&3&1&4\\0&0&1&-1&2&1\\0&0&0&0&0&-4\end{bmatrix}\to\begin{bmatrix}\boxed{1}&2&0&5&-3&0\\0&0&\boxed{1}&-1&2&0\\0&0&0&0&0&\boxed{1}\end{bmatrix}$

$U=\begin{bmatrix}\boxed{1}&2&0&5&-3&0\\0&0&\boxed{1}&-1&2&0\\0&0&0&0&0&\boxed{1}\end{bmatrix}$

a basis for the $col(A)=\left\{\begin{bmatrix}1\\2\\3\end{bmatrix},\begin{bmatrix}2\\5\\7\end{bmatrix},\begin{bmatrix}4\\9\\9\end{bmatrix}\right\}$

dependent variables of $U=\left\{\begin{bmatrix}2\\4\\6\end{bmatrix},\begin{bmatrix}3\\5\\8\end{bmatrix},\begin{bmatrix}1\\4\\5\end{bmatrix}\right\}$

$\begin{bmatrix}2\\4\\6\end{bmatrix}=2\begin{bmatrix}1\\2\\3\end{bmatrix}$, $\begin{bmatrix}3\\5\\8\end{bmatrix}=5\begin{bmatrix}1\\2\\3\end{bmatrix}-\begin{bmatrix}2\\5\\7\end{bmatrix}$, $\begin{bmatrix}1\\4\\5\end{bmatrix}=-3\begin{bmatrix}1\\2\\3\end{bmatrix}+2\begin{bmatrix}2\\5\\7\end{bmatrix}$

14. $U=[u_1,u_2,u_3,u_4]$

$u_3=2u_1+u_2$, $u_4=u_1+4u_2$

$\Rightarrow a_3=2a_1+a_2=2\begin{bmatrix}3\\5\\3\\1\end{bmatrix}+\begin{bmatrix}4\\-3\\-1\\-1\end{bmatrix}=\begin{bmatrix}-2\\7\\1\\1\end{bmatrix}$

$a_4=a_1+4a_2=\begin{bmatrix}-3\\5\\3\\1\end{bmatrix}+4\begin{bmatrix}4\\-3\\-1\\-1\end{bmatrix}=\begin{bmatrix}13\\-7\\-1\\-3\end{bmatrix}$

15. $x=Null(A)=\alpha z_1+\beta z_2=\begin{bmatrix}\alpha+\beta\\ \alpha\\ 2\alpha-\beta\end{bmatrix}$

$Ax=0 \Rightarrow a_1(\alpha+\beta)+a_2\alpha+a_3(2\alpha-\beta)=0 \quad \therefore \alpha(a_1+a_2+2a_3)+\beta(a_1-a_3)=0$

$\therefore b=a_1+2a_2+a_3 \quad \therefore b=a_2-a_3+\frac{\beta}{\alpha}(a_3-a_1)$

$Ax=b \quad [a_1\ a_2\ a_3][x_1\ x_2\ x_3]^T=-\frac{\beta}{\alpha}a_1+a_2+(\frac{\beta}{\alpha}-1)a_3$

$\therefore x=[-\frac{\beta}{\alpha}\quad 1\quad \frac{\beta}{\alpha}-1]^T=[-k\quad 1\quad k-1]^T \quad (k\neq 0, k\in R)$

16. ① ② ④ ⑦

17. $A=\left[\begin{array}{cccc|c}-2&4&-2&-4&0\\2&-6&-3&1&0\\-3&8&2&-3&0\end{array}\right]\to\left[\begin{array}{cccc|c}-1&2&-1&-2&0\\0&-2&-5&-3&0\\0&2&5&3&0\end{array}\right]\to\left[\begin{array}{cccc|c}-1&2&-1&-2&0\\0&-2&-5&-3&0\\0&0&0&0&0\end{array}\right]\to\left[\begin{array}{cccc|c}1&0&6&5&0\\0&1&\frac{5}{2}&\frac{3}{2}&0\\0&0&0&0&0\end{array}\right]$

$\therefore \begin{cases}x_1=-6x_3-5x_4\\ x_2=-\frac{5}{2}x_3-\frac{3}{2}x_4\end{cases} \quad \therefore Null(A)=span\left\{\begin{bmatrix}-6\\-\frac{5}{2}\\1\\0\end{bmatrix},\begin{bmatrix}-5\\-\frac{3}{2}\\0\\1\end{bmatrix}\right\}$

$\therefore$ basis is $\left\{\begin{bmatrix}-6\\-\frac{5}{2}\\1\\0\end{bmatrix},\begin{bmatrix}-5\\-\frac{3}{2}\\0\\1\end{bmatrix}\right\}$

18. $\mathcal{E}=\{x,1\} \quad U=\{2x-1, 2x+1\}$

$2x-1=2*x-1*1$

$2x+1=2*x+1*1$

$[2x-1]_U=\begin{bmatrix}2\\-1\end{bmatrix} \quad [2x+1]_U=\begin{bmatrix}2\\1\end{bmatrix}$

Transition from U to $\mathcal{E}$ is $S=\begin{bmatrix}2&2\\-1&1\end{bmatrix}$

19. $A=\begin{bmatrix}-2&4&-2&-4\\2&-6&-3&1\\-3&8&2&-3\end{bmatrix}\to\begin{bmatrix}1&-2&1&2\\0&-2&-5&-3\\0&2&5&3\end{bmatrix}\to\begin{bmatrix}1&-2&1&2\\0&2&5&3\\0&0&0&0\end{bmatrix}=\begin{bmatrix}1&0&6&5\\0&1&\frac{5}{2}&\frac{3}{2}\\0&0&0&0\end{bmatrix}$

$\begin{cases}x_1=-6x_3-5x_4\\ x_2=-\frac{5}{2}x_3-\frac{3}{2}x_4\end{cases} \quad \therefore Null(A)=span\left(\begin{bmatrix}-6\\-\frac{5}{2}\\1\\0\end{bmatrix},\begin{bmatrix}-5\\-\frac{3}{2}\\0\\1\end{bmatrix}\right)$

$\therefore$ a basis is of $Null(A)=\left\{\begin{bmatrix}-6\\-\frac{5}{2}\\1\\0\end{bmatrix},\begin{bmatrix}-5\\-\frac{3}{2}\\0\\1\end{bmatrix}\right\}$

20. (a) $A=\begin{bmatrix}1&3&2\\2&1&4\\4&7&8\end{bmatrix}\to\begin{bmatrix}1&3&2\\0&-5&0\\0&5&0\end{bmatrix}\to\begin{bmatrix}\boxed{1}&0&2\\0&\boxed{1}&0\\0&0&0\end{bmatrix}$

a basis for $ROW(A)=\left\{\begin{bmatrix}1\\3\\2\end{bmatrix},\begin{bmatrix}2\\1\\4\end{bmatrix}\right\}$ $col(A)=\left\{\begin{bmatrix}1\\2\\4\end{bmatrix},\begin{bmatrix}3\\1\\7\end{bmatrix}\right\}$ $Null(A)=\left\{\begin{bmatrix}-2\\0\\1\end{bmatrix}\right\}$

(b) $\begin{bmatrix} -3 & 1 & 3 & 4 \\ 1 & 2 & -1 & -2 \\ -3 & 8 & 4 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -1 & -2 \\ -3 & 1 & 3 & 4 \\ -3 & 8 & 4 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & -1 & -2 \\ 0 & 7 & 0 & -2 \\ 0 & 14 & 1 & -4 \end{bmatrix} \to \begin{bmatrix} \boxed{1} & 2 & 0 & -2 \\ 0 & \boxed{1} & 0 & -\frac{2}{7} \\ 0 & 0 & \boxed{1} & 0 \end{bmatrix}$

$\to \begin{bmatrix} 1 & 0 & 0 & -\frac{10}{7} \\ 0 & 1 & 0 & -\frac{2}{7} \\ 0 & 0 & 1 & 0 \end{bmatrix}$ ∴ a basis for $COL(A) = \left\{ \begin{bmatrix} -3 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 8 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ 4 \end{bmatrix} \right\}$

a basis for $Null(A) = \left\{ \begin{bmatrix} \frac{10}{7} \\ \frac{2}{7} \\ 0 \\ 1 \end{bmatrix} \right\}$ a basis for $row(A) = \left\{ \begin{bmatrix} -3 \\ 1 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -1 \\ -2 \end{bmatrix}, \begin{bmatrix} -3 \\ 8 \\ 4 \\ 2 \end{bmatrix} \right\}$

(c) $\begin{bmatrix} 1 & 3 & -2 & 1 \\ 2 & 1 & 3 & 2 \\ 3 & 4 & 5 & 6 \end{bmatrix} \to \begin{bmatrix} 1 & 3 & -2 & 1 \\ 0 & -5 & 7 & 0 \\ 0 & 0 & 4 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & 3 & -2 & 1 \\ 0 & 1 & -\frac{7}{5} & 0 \\ 0 & 0 & 1 & \frac{3}{4} \end{bmatrix} \to \begin{bmatrix} \boxed{1} & 0 & 0 & -\frac{13}{20} \\ 0 & \boxed{1} & 0 & \frac{21}{20} \\ 0 & 0 & \boxed{1} & \frac{3}{4} \end{bmatrix}$

∴ a basis for $COL(A) = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ 5 \end{bmatrix} \right\}$

a basis for $Null(A) = \left\{ \begin{bmatrix} \frac{13}{20} \\ -\frac{21}{20} \\ -\frac{3}{4} \\ 1 \end{bmatrix} \right\}$

a basis for $row(A) = \left\{ \begin{bmatrix} 1 \\ 3 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 5 \\ 6 \end{bmatrix} \right\}$